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THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

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OMB NO. 0704-0188

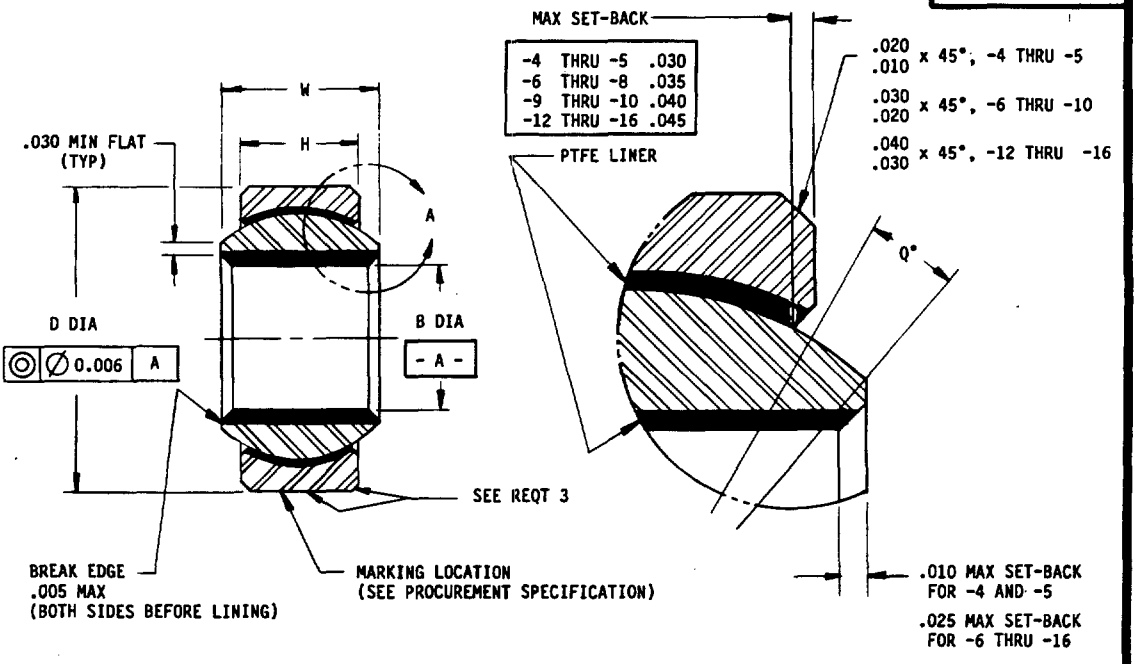


TABLE I. DIMENSIONS AND STRENGTHS

DASH NO.	B +.0000 -.0010	D +.0000 -.0005	H ±.005	Q MIN	W +.000 -.002	STATIC LIMIT LOAD		OSCILLATING LOAD LB	NO-LOAD ROTATIONAL BREAKAWAY TORQUE IN - LB		WT LB MAX
						RADIAL LB	AXIAL LB		STANDARD	* "K" TYPE	
-4	.2510	.6562	.250	10	.343	5550	430	2650	1.0 - 5.0	0 - 0.5	.020
-5	.3135	.7500	.281	10	.375	7700	700	3700			.030
-6	.3760	.8125	.312	9	.406	10200	1100	4900			.040
-7	.4385	.9062	.343	8	.437	12950	1400	6700			.050
-8	.5010	1.0000	.390	8	.500	17250	2100	8250	1.0 - 15.0	0 - 1.0	.070
-9	.5635	1.0937	.437	8	.562	22150	3680	10600			.090
-10	.6260	1.1875	.500	8	.625	27700	4720	13250			.120
-12	.7510	1.4375	.593	8	.750	40600	6750	19400			.210
-14	.8760	1.5625	.703	8	.875	55950	9350	26750	1.0 - 25.0	0 - 2.0	.270
-16	1.0010	1.7500	.797	9	1.000	73800	12160	35250			.390

* SEE REQUIREMENT 5 "NO-LOAD TORQUE" AND NOTE 5

REQUIREMENTS:

1. MATERIAL: BALL: PH13-8 Mo STEEL ALLOY PER AMS 5629, CONDITION H1000.
   OUTER RING: 17-4 PH STEEL ALLOY PER AMS 5643.
   LINER: POLYTETRAFLUOROETHYLENE (PTFE) SHALL BE INCLUDED IN THE LINER.

2. SURFACE TEXTURE: BALL DIA. Ra 8; BALL FACES Ra 16; OUTER RACE DIA Ra 32; ALL OTHER METALLIC SURFACES Ra 125. LINER SURFACES ARE EXEMPT FROM SURFACE TEXTURE MEASUREMENTS.

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY —
AIR FORCE — 99 DLA —
REVIEW: DLA - IS
USER: ARMY - MI
PROJECT NUMBER: 3120 - 0695

MILITARY SPECIFICATION SHEET

TITLE
BEARING, PLAIN, SELF-ALIGNING, SELF-LUBRICATING, LINED BORE, LOW SPEED, NARROW, CHAMFERED OUTER RING, -65° TO 325° F

SPECIFICATION SHEET NUMBER 31 JAN 90
MIL-B-81820/4

SUPERSEDING

AMSC — N/A FSC 3120

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3. SURFACE FINISH:

OUTER RACE: PLATING, WHEN SPECIFIED, SHALL BE CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2.

PLATE ON THE OUTSIDE DIAMETER SURFACE AND ON THE CHAMFER.

PLATING RUNOUT SHALL OCCUR IN THE SIDE FACE AREA BETWEEN THE CHAMFER AND THE BALL.

THE PTFE LINER IN THE OUTER RACE INSIDE DIAMETER AND IN THE BALL BORE SHALL BE PROTECTED FROM EXPOSURE TO PLATING SOLUTIONS DURING PROCESSING.

BALL: PASSIVATE PER MIL-S-5002.

4. HARDNESS: BALL: Rc 43 MIN; OUTER RING: Rc 28 MIN, Rc 37 MAX BEFORE SWAGING.

5. NO-LOAD TORQUE: WHEN THE LETTER "K" IS PRESENT IN THE PART NUMBER LOWER VALUES OF NO-LOAD TORQUE ARE SPECIFIED PER TABLE I. IF THE MEASURED TORQUE OF A "K" TYPE BEARING IS LESS THAN 0.1 INCH-POUND. THE INTERNAL RADIAL CLEARANCE SHALL BE MEASURED AND SHALL NOT EXCEED THE VALUES IN TABLE II. THESE REQUIREMENTS APPLY TO THE TORQUE AND INTERNAL PLAY BETWEEN THE SPHERICAL BALL AND THE OUTER RING. THIS STANDARD DOES NOT DEFINE REQUIREMENTS FOR TORQUE OR INTERNAL PLAY BETWEEN THE BEARING AND SHAFT.

TABLE II. INTERNAL CLEARANCE

DASH NO.	MAXIMUM RADIAL PLAY	MAXIMUM AXIAL PLAY
-4K THRU -12K	0.0007 INCH	0.0028 INCH
-14K THRU -16K	0.0010 INCH	0.0040 INCH

6. BREAK SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

7. DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES, DECIMALS ± .010, ANGLES ± 0.5.

NOTES:

1. WHEN TESTED TO THE FLUID CONTAMINATION AND SUB-ZERO TEMPERATURE REQUIREMENTS OF THE PROCUREMENT SPECIFICATION, THE OSCILLATING LOAD SHALL BE DECREASED TO 75% OF SPECIFIED LOAD.

2. WHEN FLUIDS AND ELEVATED TEMPERATURES (ABOVE 200°F) ARE BOTH PRESENT IN AN APPLICATION, THEN REDUCTIONS IN OPERATING LOADS OR BEARING LIFE MAY BE REQUIRED.

3. DASH NUMBER DESIGNATES NOMINAL BORE IN SIXTEENTHS OF AN INCH.

4. DIMENSION "D" TO BE MET AFTER PLATING.

5. EXAMPLE OF PART NO.

M81820/4-6 P K = 3/8 NOMINAL BORE DIA, PH13-8 Mo BALL MATERIAL, CADMIUM PLATED, LOW TORQUE.

- K: ABSENCE OF LETTER INDICATES STANDARD BREAKAWAY TORQUE.
- K: LETTER "K" INDICATES LOW BREAKAWAY TORQUE (SEE TABLE I).
- P: ABSENCE OF LETTER INDICATES NO PLATING.
- P: LETTER "P" INDICATES CADMIUM PLATING (SEE REQUIREMENT 3).
- 6: SIZE - BORE DIAMETER. (DASH NO. FROM TABLE I).
- M81820/4: MS NUMBER.

6. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

7. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY —
AIR FORCE — 99 DLA —
REVIEW: DLA-IS
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